

# CHAPTER 7

7-1. Char. eq. :  $(z-p_1)^2 (z-p_{n+1}) \cdots (z-p_n) = 0$

In (7-1):

$$C(z) = \frac{k_1 z}{z-p_1} + \cdots + \frac{k_n f_i(z)}{(z-p_i)^2} + \frac{k_{n+1} z}{z-p_{n+1}} + \cdots + \frac{k_n z}{z-p_n}$$

where, from the table in the appendix,

$$\mathcal{Z}^{-1} \left[ \frac{k_i f_i(z)}{(z-p_i)^i} \right] = k_i (k)^{i-1} p_i^k$$

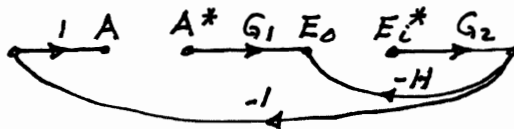
$$\therefore c(k) = k_1 p_1^k + k_2 k (p_2^k) + \cdots + k_n (p)^{n-1} (p_i^k) + \cdots$$

Now

$$\begin{aligned} \lim_{k \rightarrow \infty} k^{i-1} p_i^k &= \lim_{k \rightarrow \infty} \frac{k^{i-1}}{p_i^{-k}} = \lim_{k \rightarrow \infty} \frac{(i-1)k^{i-2}}{-k(p_i)^{k-1}} \\ &= \cdots = \lim_{k \rightarrow \infty} \frac{(i-1)(i-2) \cdots k^0}{-k(\quad)} \end{aligned}$$

The numerator is constant and the denominator is unbounded for  $|p_i| < 1$ . Thus all terms in  $c(k)$  of the form  $(k)^{i-1} (p_i)^k \rightarrow 0$  for  $|p_i| < 1$

7-2.



$$A = -G_2 E_i^* \Rightarrow A^* = -G_2^* E_i^*$$

$$E_0 = G_1 A^* - G_2 H E_i^* \Rightarrow E_0^* = -G_1^* G_2^* E_i^* - \overline{G_2 H}^* E_i^*$$

$$\text{Let } E_0^* = E_i^*$$

$$\therefore (1 + G_1^* G_2^* + \overline{G_2 H}^*) E_i^* = 0$$

$$\therefore \underline{1 + G_1(z) G_2(z) + \overline{G_2 H}(z) = 0}$$

$$7-3.(a) \quad T(z) = \frac{N(z)}{D(z)} = \frac{N(z)}{(z-z_1)(z-z_2) \cdots (z-z_n)}$$

$$7-3. (a) \frac{C(z)}{z} = \frac{N(z)}{(z-1)(z-z_1)\dots(z-z_n)} = \frac{k_0}{z-1} + \frac{k_1}{z-z_1} + \dots + \frac{k_n}{z-z_n}$$

$$\therefore C(kT) = k_0 + k_1(z_1)^k + \dots + k_n(z_n)^k, \quad |z_i| < 1, \quad i=1,2,\dots,n$$

$$\therefore \lim_{k \rightarrow \infty} C(kT) = \underline{k_0}$$

$$(b) T(z) = \frac{N(z)}{D(z)}, \quad \therefore N(z) = (z-1)N_1(z) \quad [T(1) = 0]$$

$$(c) R(z) = 1$$

$$\frac{C(z)}{z} = \frac{k_1}{z-z_1} + \frac{k_2}{z-z_2} + \dots + \frac{k_n}{z-z_n}$$

$$\therefore C(kT) = k_1(z_1)^k + \dots + k_n(z_n)^k$$

$$\therefore \lim_{k \rightarrow \infty} C(kT) = 0, \quad |z_i| < 1$$

$$7-4. T=0.5, \text{ char. eq. : } (z-0.9)(z-0.8)(z^2-1.9z+1)=0$$

$$\text{complex roots: } z = 0.95 \pm j0.3122 = 1 \angle \pm 18.19^\circ = 1 \angle \pm 0.3175 \text{ rad}$$

$$(a) k_1(0.9)^k, k_2(0.8)^k, A \cos(0.3175k + \theta)$$

$$(b) \text{ marginally stable}$$

$$(c) \text{ root} = 1 \angle \pm \omega T, \therefore \omega T = 0.3175 \Rightarrow \omega = \frac{0.3175}{0.5} = \underline{0.635} \text{ rad/s}$$

$$7-5. (a) \text{ (i) } Q(1) = 1.3 - 1.1 > 0 \checkmark$$

$$(-1)^2 Q(-1) = 1.3 + 1.1 > 0 \checkmark$$

$$|a_0| < a_2 \Rightarrow 0.3 < 1 \checkmark$$

$$\therefore \underline{\text{stable}}$$

$$\text{ (ii) } Q(1) = 1.25 - 1 > 0 \checkmark$$

$$(-1)^2 Q(-1) = 1 + 0.25 > 0 \checkmark$$

$$|a_0| < a_2 \Rightarrow 0.25 < 1 \checkmark$$

$$\therefore \underline{\text{stable}}$$

$$\text{ (iii) } Q(1) = 0.7 - 0.1 > 0 \checkmark$$

$$(-1)^2 Q(-1) = 1 + 0.1 - 0.3 > 0 \checkmark$$

$$|a_0| < a_2 \Rightarrow 0.3 < 1 \checkmark$$

$$\therefore \underline{\text{stable}}$$

$$\text{ (iv) } Q(1) = 1 - 0.25 > 0 \checkmark$$

$$(-1)^2 Q(-1) = 1 - 0.25 > 0 \checkmark$$

$$|a_0| < a_2 \Rightarrow 0.25 < 1 \checkmark$$

$$\therefore \underline{\text{stable}}$$

$$7-5. (iv) Q(1) = 1 - 1.6 + 1 > 0^{\checkmark}$$

$$(-1)^2 Q(-1) = 1 + 1.6 + 1 > 0^{\checkmark}$$

$$|a_0| < a_2 \Rightarrow 1 < 1^{\checkmark}$$

$\therefore$  not stable

$$(vii) Q(1) = 1 - 2.2 + 1.55 - 0.35 = 0 \therefore \text{not stable}$$

$$(viii) Q(1) = 1 - 1.9 + 1.4 - 0.45 > 0^{\checkmark}$$

$$(-1)^3 Q(-1) = -1(-1 - 1.9 - 1.4 - 0.45) > 0^{\checkmark}$$

$$|a_0| < a_3 \Rightarrow 0.45 < 1$$

| $z^0$  | $z^1$ | $z^2$  | $z^3$ |
|--------|-------|--------|-------|
| -0.45  | 1.4   | -1.9   | 1     |
| 1      | -1.9  | 1.4    | -0.45 |
| -0.798 | 1.27  | -0.545 |       |

$$b_0 = \begin{vmatrix} -0.45 & 1 \\ 1 & -0.45 \end{vmatrix} = -0.798$$

$$b_1 = \begin{vmatrix} -0.45 & -1.9 \\ 1 & 1.4 \end{vmatrix} = 1.27$$

$$b_2 = \begin{vmatrix} -0.45 & 1.4 \\ 1 & -1.9 \end{vmatrix} = -0.545$$

$$|b_0| > |b_2| \Rightarrow 0.798 > 0.545$$

$\therefore$  stable

$$(b) (i) \text{ zeros: } 0.5, 0.6 \Rightarrow k_1(0.5)^n, k_2(0.6)^n$$

$$(ii) \text{ zeros: } 0.5, 0.5 \Rightarrow k_1(0.5)^n, k_2 n(0.5)^n$$

$$(iii) \text{ zeros: } -0.5, 0.6 \Rightarrow k_1(-0.5)^n, k_2(0.6)^n$$

$$(iv) \text{ zeros: } -0.5, 0.5 \Rightarrow k_1(-0.5)^n, k_2(0.5)^n$$

$$(v) \text{ zeros: } 1/\pm 0.643 \text{ rad} \Rightarrow A \cos(0.643n + \theta)$$

$$(vi) \text{ zeros: } 0.9, 1.1 \Rightarrow k_1(0.9)^n, k_2(1.1)^n$$

$$(vii) \text{ zeros: } 1, 0.7, 0.5 \Rightarrow k_1, k_2(0.7)^n, k_3(0.5)^n$$

$$(viii) \text{ zeros: } 0.9, 0.707/\pm \pi/4 \Rightarrow k_1(0.9)^n, A(0.707)^n \cos(\pi/4 n + \theta)$$

$$(c) (v) A \cos(0.643n + \theta), \text{ marginally stable}$$

$$(vi) k_2(1.1)^n, \text{ unstable}$$

$$(vii) k_1(1)^n = k_1, \text{ marginally stable}$$

$$7-6. (a) G(z) = \frac{z-1}{z} \mathcal{Z}\left[\frac{1}{s(s+1)}\right] = \frac{z-1}{z} \frac{(1-e^{-T})z}{(z-1)(z-e^{-T})} \Rightarrow \frac{0.6321}{z-0.3678}$$

$$7-6. (a) \therefore 1 + KG(z) = 0 = z - 0.3679 + 0.6321K$$

$$(b) \quad z = 0.3679 - 0.6321K > -1 \Rightarrow K < \frac{1.3679}{0.6321} = \underline{2.164}$$

$$z = 0.3679 - 0.6321K < 1 \Rightarrow K > \underline{-1}$$

$$\therefore \underline{-1 < K < 2.164}$$

$$(c) \quad K = -1, \quad z = 1 \Rightarrow \underline{k(1)^n = k}$$

$$(d) \quad K = 2.164, \quad z = -1 \Rightarrow \underline{k(-1)^n}$$

$$(e) \quad \begin{array}{l} K = -1; \\ \text{for } k=1:2 \\ \quad q = [1 \quad (-0.3679 + 0.6321*K)]; \\ \quad \text{roots}(q) \\ \quad K = 2.164; \\ \text{end} \end{array}$$

$$7-7. (a) \text{ From Problem 7-6: } G(z) = \frac{0.6321}{z - 0.3679}$$

$$\therefore \text{char. eq. : } \underline{z - 0.3679 + 0.6321K = 0}$$

$$(b) \quad z = \frac{1 + \frac{T}{2}w}{1 - \frac{T}{2}w} = \frac{1 + 0.5W}{1 - 0.5W} = \frac{z + W}{z - W}$$

$$\therefore \text{char. eq. : } z + W + (z - W)(-0.3679 + 0.6321K) = 0$$

$$\therefore (1.3679 - 0.6321K)W + (1.2642 + 1.2642K) = 0$$

$$\therefore \begin{array}{l|l} W^1 & 1.3679 - 0.6321K \Rightarrow K < 2.164 \\ W^0 & 1.2642 + 1.2642K \Rightarrow K > -1 \end{array}$$

$$\therefore \underline{-1 < K < 2.164}$$

$$(c) \quad Q(1) > 0 \Rightarrow 1 - 0.3679 + 0.6321K > 0 \Rightarrow K > -1$$

$$(-1)Q(-1) > 1 \Rightarrow 1 + 0.3679 - 0.6321K > 0 \Rightarrow K > -2.164$$

$$|a_0| < 1 \Rightarrow |-0.3679 + 0.6321K| < 1 \Rightarrow -1 < K < 2.164$$

$$\therefore \underline{-1 < K < 2.164}$$

$$(d) \quad \underline{K = 2.164} : w = \frac{1.2642 + 1.2642K}{1.3679 - 0.6321K} = \frac{(\quad)}{1.3679 - 1.3679} \therefore \underline{\text{unbounded}}$$

$$z = 0.3679 - 0.6321(2.164) = -1 = 1 \angle \pi = 1 \angle \omega T$$

$$\therefore \omega = \pi = \underline{3.1416 \text{ rad/s}}$$

$$7-7. (e) \quad 1 + K G_p(s) = 1 + \frac{K}{s+1} \Rightarrow s+1+K=0$$

$$\therefore s = -1-K \Rightarrow \underline{K > -1}$$

(f) Sampling destabilizes this system.

(g)

$$\begin{aligned} K &= -1; \\ w_0 &= 1.2642 + 1.2642*K \\ K &= 2.164; \\ w_1 &= 1.3679 - 0.6321*K \end{aligned}$$

7-8. (a) From Problem 7-6,  $z - e^{-T} + (1 - e^{-T})K = 0$ , char. eq.

(b) From Problem 7-6,  $T=1 \Rightarrow \underline{K < 2.164}$

$$T=0.1 \Rightarrow z - 0.9048 + 0.09516K = 0$$

$$\therefore z = 0.9048 - 0.09516K > -1 = \underline{K < 20.02}$$

$$T=0.01 \Rightarrow z - 0.99005 + 0.00995K = 0$$

$$\therefore z = 0.99005 - 0.00995K > -1 \Rightarrow \underline{K < 200.0}$$

(c) From Problem 7-7,  $K < \infty$

(d) Reducing  $T$  improves stability for this system.

(e)

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T = 0.1;
K = 20.02;
for k=1:2
    q = [1 (-exp(-T) + (1 - exp(-T))*K)]
    roots(q)
    T = .01;
    K = 200.0;
end
    
```

$$7-9. (a) \quad 1 + K G(z) H = 0 = 1 + \frac{(1.037)K}{z-0.7408} (0.04) = 0$$

$$\therefore \text{char. eq.: } \underline{Q(z) = z - 0.7408 + 0.04148K = 0}$$

$$(b) \quad G(w) = \frac{1.037}{\frac{1+0.3w}{1-0.3w} - 0.7408} = \frac{-0.5958w + 1.986}{w + 0.4964}$$

$$1 + K G(w) H = 0 = 1 + \frac{0.04K(-0.5958w + 1.986)}{w + 0.4964}$$

$$\therefore \text{char. eq.: } \underline{(1 - 0.02383K)w + 0.4964 + 0.07944K = 0}$$

$$7-9. (b) \therefore w^1 \left| \begin{array}{l} 1 - 0.02383K \\ 0.4964 + 0.07944K \end{array} \right. \Rightarrow K < 41.96$$

$$w^0 \left| \begin{array}{l} 1 - 0.02383K \\ 0.4964 + 0.07944K \end{array} \right. \Rightarrow K > -6.249$$

$$\therefore \text{stable: } -6.249 < K < 41.96$$

$$(c) \text{ From (a): } \frac{z^1}{1} \quad \frac{z^0}{0.04148K - 0.7408}$$

$$Q(1) = 1 - 0.7408 + 0.04148K > 0 \Rightarrow K > -6.249$$

$$(-1)Q(1) = 1 + 0.7408 - 0.04148K > 0 \Rightarrow K < 41.96$$

$$|a_0| < |a_1|: |0.04148K - 0.7408K| < 1 \Rightarrow \text{same as above}$$

$$\therefore -6.249 < K < 41.96$$

$$(d) G(z) = \frac{z-1}{z} \mathcal{Z} \left[ \frac{z}{s(s+0.5)} \right] = \frac{z-1}{z} \left( \frac{z}{0.5} \right) \left[ \frac{z(1-e^{-0.03})}{(z-1)(z-e^{-0.03})} \right] = \frac{0.1182}{z-0.9704}$$

$$1 + KG(z)H = 1 + \frac{0.1182K}{z-0.9704} (0.04) = 0$$

$$\therefore Q(z) = z - 0.9704 + 0.004729K = 0$$

$$Q(1) = 1 - 0.9704 + 0.004729K > 0 \Rightarrow K < -6.259$$

$$-Q(-1) = 1 + 0.9704 - 0.004729K > 0 \Rightarrow K > 416.7$$

$$\therefore -6.259 < K < 416.7$$

(e) As  $T \rightarrow 0$ , the range of  $K$  for stability becomes larger.  
Thus sampling tends to destabilize this system.

$$(f) \quad T = 0.6;$$

$$K = -6.249;$$

$$q = [1 \quad (-\exp(-T*0.5)) + (1 - \exp(-T*0.5)*K*0.04*4)]$$

$$\text{roots}(q)$$

$$7-10. (a) 1 + KG(z)H = 0 = z^2 - 1.8187z + 0.8187 + 0.001311Kz + 0.001226K$$

$$\therefore \text{char. eq.: } Q(z) = z^2 - (1.8187 - 0.001311K)z + 0.8187 + 0.001226K = 0$$

$$(b) z = \frac{1 + \frac{T}{2}W}{1 - \frac{T}{2}W} = \frac{1 + 0.05W}{1 - 0.05W} = \frac{20+W}{20-W}$$

$$\therefore Q(W) = (20+W)^2 - (1.8187 - 0.001311K)(400-W^2) + (0.8187 + 0.001226K)(20-W)^2 = 0$$

$$\therefore (3.6374 - 0.00085K)W^2 + (7.252 - 0.04904K)W + 1.0148K = 0$$

$$\begin{array}{lcl}
 7-10. (b) & w^2 & 3.6374 - 0.000085K \quad 1.0148K \Rightarrow K < 42,793 \\
 & w^1 & 7.252 - 0.04904K \Rightarrow K < 147.9 \\
 & w^0 & 1.0148K \Rightarrow K > 0
 \end{array}$$

$\therefore$  for stability,  $0 < K < 147.9$

$$\begin{aligned}
 (c) \quad Q(1) > 0 &\Rightarrow 1 - 1.8187 + 0.001311K + 0.8187 + 0.001226K > 0 \\
 (1)^2 Q(-1) > 0 &\Rightarrow 1 + 1.8187 - 0.001311K + 0.8187 + 0.001226K > 0 \quad \checkmark \because K > 0 \\
 &\therefore K < 42,793
 \end{aligned}$$

$$|a_0| < a_2 \Rightarrow 0.8187 + 0.001226K < 1, \therefore K < 147.9$$

For Stability:  $0 < K < 147.9$

$$(d) \quad K = 147.9 \quad \text{From (a): char. eq. } z^2 - 1.6248z + 1 = 0$$

$$\text{zeros: } z = 1 \angle \pm 35.67^\circ = 1 \angle \pm 0.6225 \text{ rad.}$$

$$\text{From (b): char. eq. } 3.6148w^2 + 150.09 = 0$$

$$\therefore w = \pm j 6.437$$

$$(e) \quad z\text{-plane: } 1/\omega T = 1/0.6225 \quad \therefore \omega = \frac{0.6225}{0.1} = 6.225 \text{ rad/s}$$

$$w\text{-plane: } \omega_w = 6.437$$

$$(f) \quad (7-10) \quad \omega_w = \frac{2}{T} \tan\left(\frac{\omega T}{2}\right) = \frac{2}{0.1} \tan\left(\frac{0.6225}{2}\right) = 6.433^\circ$$

$$\begin{aligned}
 (g) \quad K &= 147.9; \\
 q &= [1 \quad (-1.8187 + 0.001311K) \quad (0.8187 + 0.001226K)] \\
 \text{roots}(q)
 \end{aligned}$$

$$7-11. (a) \quad 1 + KG(z)H = 0 = z^2 - 1.7408z + 0.7408 + 0.009072Kz +$$

$$\therefore \text{char. eq.: } z^2 + (0.009072K - 1.7408)z + 0.7408 + 0.008208K = 0$$

$$(b) \quad z = \frac{1 \pm \frac{1}{2}W}{1 - \frac{1}{2}W} = \frac{1 + 0.025W}{1 - 0.025W} = \frac{40 + W}{40 - W} \text{ yields}$$

$$(40 + W)^2 + (0.009072K - 1.7408)(1600 - W^2) + (0.7408 + 0.008208K)(40 - W)^2 = 0$$

$$\therefore (3.4816 - 0.000864K)W^2 + (20.736 - 0.65664K)W + 27.65K = 0$$

$$w^2 \quad 3.4816 - 0.000864K \quad 27.65K \Rightarrow K < 4,029.6$$

$$w^1 \quad 20.736 - 0.65664K \Rightarrow K < 31.58$$

$$w^0 \quad 27.65K \Rightarrow K > 0$$

$\therefore$  for stability:  $0 < K < 31.58$

$$7-11. (c) Q(1) > 0 \Rightarrow 1 + 0.009072K - 1.7408 + 0.7408 + 0.008208K > 0 \Rightarrow \underline{K > 0}$$

$$(-1)^2 Q(-1) > 0 \Rightarrow 1 - 0.009072K + 1.7408 + 0.7408 + 0.008208K > 0 \Rightarrow \underline{K < 4029.6}$$

$$|a_0| < a_2 \Rightarrow 0.7408 + 0.008208K < 1 \Rightarrow \underline{K < 31.58}$$

$\therefore$  for stability:  $\underline{0 < K < 31.58}$

$$(d) K = 31.58, \text{ From (a): } z^2 - 1.4543z + 1 = 0$$

$$\therefore z = 1/\pm 43.36^\circ = 1/\pm 0.7567 \text{ rad}$$

$$\text{From (b): } 3.4543w^2 + 873.2 = 0 \Rightarrow w = \pm j/5.90 = \pm j\omega_w$$

$$(e) z\text{-plane: } 1/\omega T = 1/0.7567 \Rightarrow \omega = 0.7567/0.05 = \underline{15.13 \text{ rad/s}}$$

$$w\text{-plane: } \omega_w = \underline{15.90}$$

$$(f) (7-10) \omega_w = \frac{z}{T} \tan\left(\frac{\omega T}{2}\right) = 40 \tan\left(\frac{0.7567}{2}\right) = \underline{15.90}$$

$$(g) \begin{array}{l} K = 31.58; \\ q = [1 \quad (0.009072K - 1.7408) \quad (0.7408 + 0.008208K)] \\ \text{roots}(q) \end{array}$$

$$7-12. (a) 1 + KG(z)H = z^2 - 2z + 1 + 0.001Kz + 0.001K = 0$$

$$\therefore \text{char. eq.: } \underline{z^2 + (0.001K - 2)z + (0.001K + 1) = 0}$$

$$(b) z = \frac{\frac{z}{T} + w}{\frac{z}{T} - w} = \frac{20 + w}{20 - w}$$

$$\therefore (20 + w)^2 + (0.001K - 2)(400 - w^2) + (0.001K + 1)(20 - w)^2 = 0$$

$$\therefore \underline{4w^2 - 0.04Kw + 0.08K = 0}$$

$$w^2 \mid \begin{array}{cc} 4 & 0.08K \end{array}$$

$$w^1 \mid -0.04K$$

$$w^0 \mid 0.08K$$

$$\Rightarrow \underline{K < 0}$$

$$\Rightarrow \underline{K > 0}$$

$\therefore$  unstable for all K

$$(c) Q(1) > 0 \Rightarrow 1 + 0.001K - 2 + 0.001K + 1 > 0 \Rightarrow \underline{K > 0}$$

$$(-1)^2 Q(-1) > 0 \Rightarrow 1 + 2 - 0.001K + 0.001K + 1 = 4 > 0$$

$$|a_0| < a_2 \Rightarrow 0.001K + 1 < 1 \Rightarrow \underline{K < 0}$$

$\therefore$  unstable for all K



$$7-13. (a) \quad 1 + G(z) = z^2 - 1.6z + 0.6 + Kz + 0.8K = 0$$

$$\therefore \text{char. eq. : } z^2 - (1.6 - K)z + 0.6 + 0.8K = 0$$

$$z = \frac{1+W}{1-W} \Rightarrow (1+W)^2 + (K-1.6)(1-W^2) + (0.6+0.8K)(1-W)^2 = 0$$

$$\therefore \text{char. eq. : } (3.2 - 0.2K)W^2 + (0.8 - 1.6K)W + 1.8K = 0$$

$$W^2 \quad \left| \begin{array}{cc} 3.2 - 0.2K & 1.8K \end{array} \right. \Rightarrow \underline{K < 16}$$

$$W^1 \quad \left| \begin{array}{cc} 0.8 - 1.6K & \end{array} \right. \Rightarrow \underline{K < 0.5}$$

$$W^0 \quad \left| \begin{array}{cc} 1.8K & \end{array} \right. \Rightarrow \underline{K > 0}$$

$$\therefore \text{for stability : } \underline{0 < K < 0.5}$$

$$(b) \quad Q(1) > 0 \Rightarrow 1 - 1.6 + K + 0.6 + 0.8K > 0 \Rightarrow \underline{K > 0}$$

$$(-1)^2 Q(-1) > 0 \Rightarrow 1 + 1.6 - K + 0.6 + 0.8K > 0 \Rightarrow \underline{K < 16}$$

$$|a_0| < a_2 \Rightarrow 0.6 + 0.8K < 1 \Rightarrow \underline{K < 0.5}$$

$$\therefore \text{for stability : } \underline{0 < K < 0.5}$$

$$(c) \quad K = 0.5; \text{ char. eq. : } 3.1W^2 + 0.9 = 0$$

$$\therefore \text{zeros: } W = \pm j 0.5388 \quad \therefore \text{marginally stable}$$

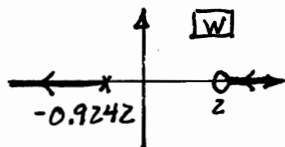
$$(d) \quad K = 0.5; \text{ char. eq. : } z^2 - 1.1z + 1 = 0$$

$$\therefore \text{zeros: } z = 0.55 \pm j 0.8352 = 1 \angle \pm 56.6^\circ, \therefore \text{marginally stable}$$

$$7-14. (a) \text{ From Problem 7-6, } G(z) = \frac{0.6321}{z - 0.3679}$$



$$(b) \quad z = \frac{z+W}{z-W} \quad \therefore \quad G(W) = \frac{0.6321(z-W)}{z+W-0.3679(z-W)} = \frac{-0.4621(W-2)}{W+0.9242}$$



$\therefore$  effectively plotting for  $K$  negative

The locus passes through infinity.

$$(c) \quad z = -1, \quad K = \left| \frac{1}{G(-1)} \right| = \left| \frac{-1 - 0.3679}{0.6321} \right| = 2.164$$

$$(d) \quad W = \infty, \quad K = \left| \frac{1}{G(\infty)} \right| = \left| \frac{1}{-0.4621} \right| = 2.164$$

7-14.(e)

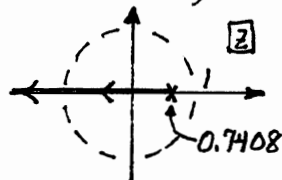
z-plane

```
k=0:0.2:5;
n = [0 0.6321];
d = [1 -0.3679];
r = rlocus(n,d,k);
[k',r]
pause
plot(real(r),imag(r),'x')
title('Root Locus')
```

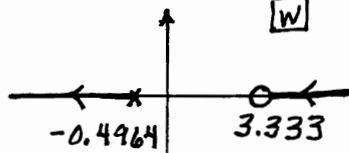
w-plane

```
k=0:0.2:5;
n = [-0.4621 0.9242];
d = [1 0.9242];
r = rlocus(n,d,k);
[k',r]
pause
plot(real(r),imag(r),'x')
title('Root Locus')
```

7-15.(a) From Problem 7-9,  $G(z) = \frac{1.037}{z - 0.7408}$



(b)  $z = \frac{1 + 0.3W}{1 - 0.3W} \Rightarrow G(W) = \frac{-0.5957(W - 3.333)}{W + 0.4964} \therefore \text{effectively plotting for } K \text{ negative.}$



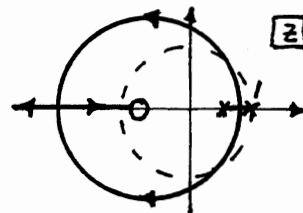
The locus passes through infinity.

(c)  $K = \left| \frac{1}{G(-1)} \right| = \frac{1.7408}{1.037} = 1.679$

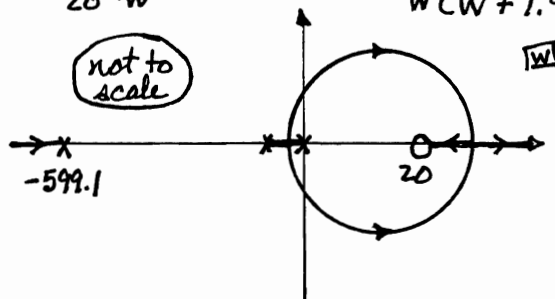
(d)  $K = \left| \frac{1}{G(\infty)} \right| = \frac{1}{0.5957} = 1.679$

(e) See Problem 7-14(e) for programs.

7-16.(a)  $G(z) = \frac{0.01873(z + 0.9354)}{(z - 1)(z - 0.8187)}$



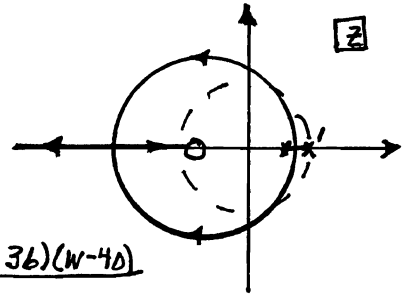
(b)  $z = \frac{20 + W}{20 - W}, G(W) = \frac{0.000333(W + 599.1)(W - 20)}{W(W + 1.994)} \therefore \text{effectively plotting for } K \text{ negative}$



Locus passes through infinity.

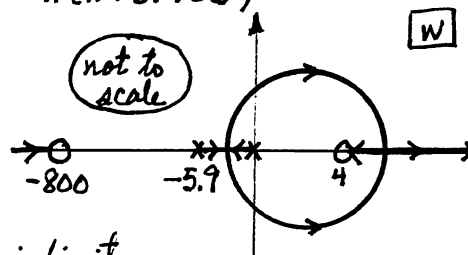
- 7-16. (c) From Problem 7-10, zeros at  $z = 1 \angle \pm 35.7^\circ$  for  $K = 147.9$   
 (d) From Problem 7-10, zeros at  $w = \pm j6.437$  for  $K = 147.9$   
 (e) See Problem 7-14(e) for programs.

7-17. (a)  $G(z) = \frac{0.02268(z + 0.9048)}{(z-1)(z-0.7408)}$



(b)  $z = \frac{40+W}{40-W}$ ;  $G(W) = \frac{0.0006202(W+800.36)(W-40)}{W(W+5.956)}$

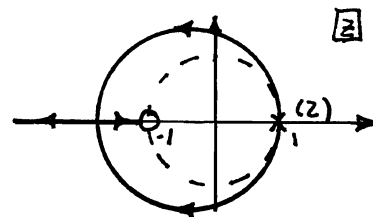
$\therefore$  effectively plotting  
for  $K$  negative



Locus passes through infinity.

- (c) From Problem 7-11, zeros at  $z = 1 \angle \pm 43.4^\circ$  for  $K = 31.58$   
 (d) From Problem 7-11, zero at  $w = \pm j15.90$  for  $K = 31.58$   
 (e) See Problem 7-14(e) for programs.

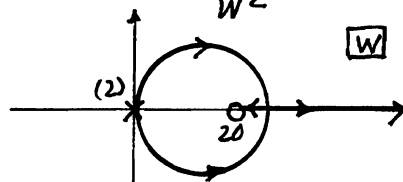
7-18. (a)  $G(z) = \frac{0.05(z+1)}{(z-1)^2}$



(b)  $z = \frac{20+W}{20-W}$

$\therefore G(W) = \frac{-0.5(W-20)}{W^2}$

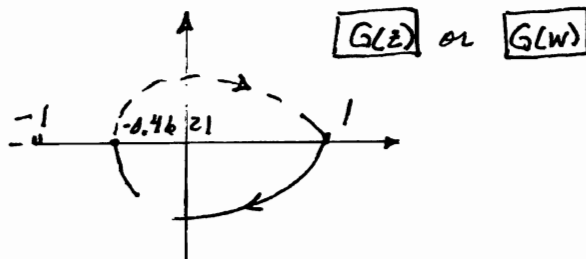
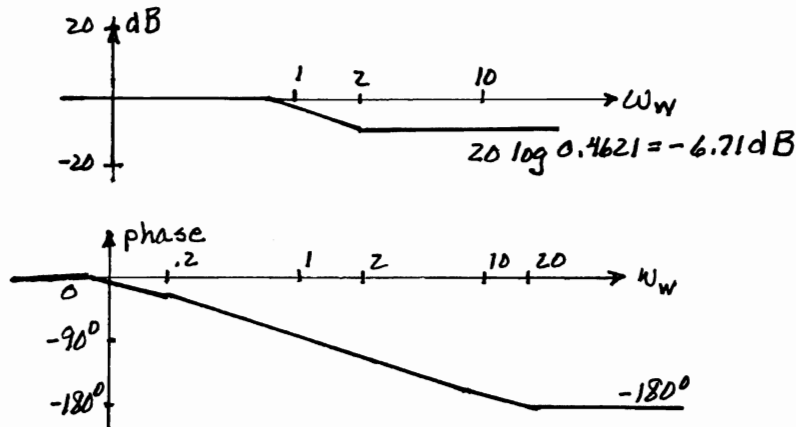
$\therefore$  effectively plotting for  $K$  negative.



- (c) Unstable for all  $K$ .  
 (d) Unstable for all  $K$ .  
 (e) See Problem 7-14(e) for programs.

7-19. (a) From Problem 7-14, stable for  $K=1$ .

$$(b) G(w) = \frac{0.4621(2-w)}{w+0.9242} = \frac{1-\frac{w}{2}}{1+\frac{w}{0.9242}}$$



(c) gain margin =  $\frac{1}{0.4621} = 2.16 \Rightarrow 6.71 \text{ dB}$   
 $\phi_m = 180^\circ$

(d)  $K = \frac{1}{0.4621} = 2.16$

(e)  $\omega_w = \infty$ ,  $z = -1 = 1/\omega T$ ,  $\therefore \omega T = \pi \Rightarrow \omega = \frac{\pi}{T} = \frac{\omega_s}{2} = 3.14 \text{ rad/s}$

(f) The problem should ask for Bode diagram.

```
num = [-0.4621 0.9242];
den = [1 0.9242];
bode(num,den)
```

7-20. (a)  $z = \frac{1+aW}{1-aW} \Rightarrow W = \frac{1}{a} \frac{z-1}{z+1}$

$\therefore W = \frac{1}{a} \frac{z-1}{z+1} \Big|_{z=e^{j\omega T}} = j \frac{1}{a} \tan\left(\frac{\omega T}{2}\right)$

(b)  $\omega_w = \frac{1}{a} \tan\left(\frac{\omega T}{2}\right)$

For  $\frac{\omega T}{2} \ll 1$ ,  $\omega_w \approx \frac{\omega T}{2a} = \frac{T}{2a} \omega$

7-20, (c)  $|z| < 1$  for stability - Let  $z = z_r + jz_i$

then  $|z|^2 = z_r^2 + z_i^2 < 1$  for stability

$$W = \frac{1}{a} \frac{z_r + jz_i - 1}{z_r + jz_i + 1} = \frac{1}{a} \frac{(z_r - 1) + jz_i}{(z_r + 1) + jz_i} \cdot \frac{(z_r + 1) - jz_i}{(z_r + 1) - jz_i}$$

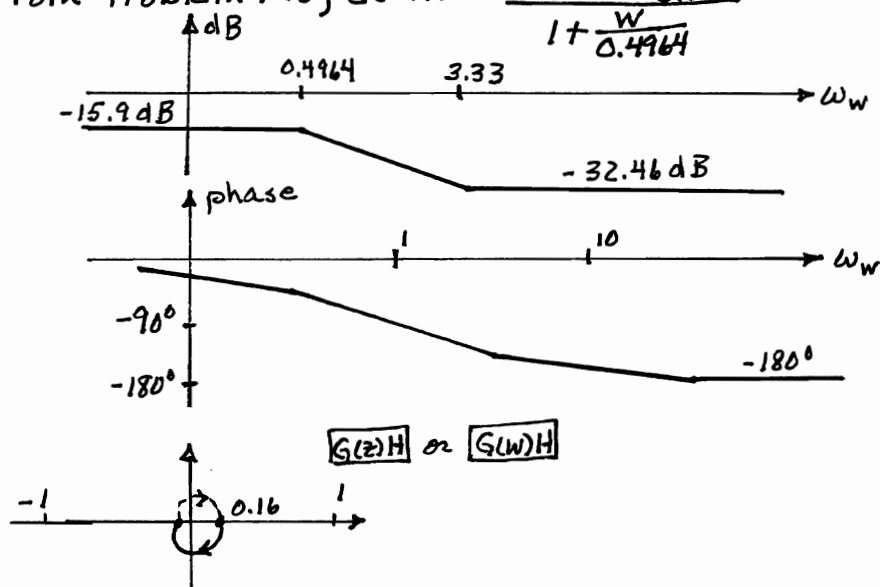
$$\text{Re}(W) = \frac{1}{a} \frac{z_r^2 - 1 + z_i^2}{(z_r + 1)^2 + z_i^2} = \frac{1}{a} \frac{|z|^2 - 1}{b}, \quad b > 0$$

$\therefore a > 0, \text{Re}(W) < 0$ , left-half plane for stability  
 $a < 0, \text{Re}(W) > 0$ , right-half plane for stability

7-21.  $\text{Output} = (5)(1.3) \cos(2t - 35^\circ) = \underline{6.5 \cos(2t - 25^\circ)}$

7-22, (a) From Problem 7-9, stable for  $K=1$ .

(b) From Problem 7-15,  $G(w)H = \frac{0.16(1 - \frac{w}{3.33})}{1 + \frac{w}{0.4964}}$



(c) gain margin =  $\frac{1}{(0.04)(0.5957)} = 41.97 \Rightarrow \underline{32.46 \text{ dB}}$

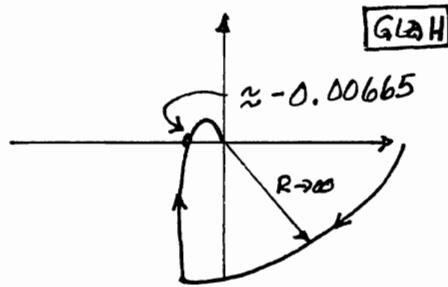
no phase margin

(d)  $K = \underline{41.97}$  from (c)

(e)  $\omega_w = \infty, z = -1 = 1 \angle 180^\circ, \therefore \omega T = \pi, \omega = \frac{\pi}{T} = \frac{\omega_s}{2} = \underline{5.24 \text{ rad/s}}$

(f) See Problem 7-19(f).

7-23.(a)



(b) From (a), stable.

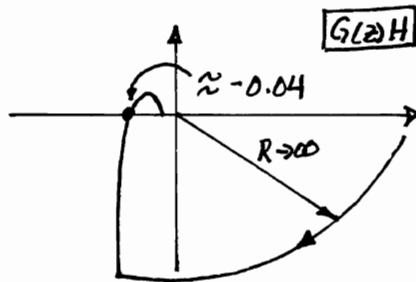
(c) Gain margin  $\approx \frac{1}{(0.095)(0.07)} = \underline{150.4}$  (147.9, from Problem 7-10)

$\omega_w = 0.15$ ,  $\text{dB} \approx 23$ ,  $\therefore \phi_m \approx 180^\circ - 95^\circ = \underline{85^\circ}$

(d) From (c),  $K \approx 150.4$

(e) From Table P7-23,  $\omega \approx \underline{6.5 \text{ rad/s}}$  with phase  $= -180^\circ$

7-24.(a)



(b) From (a), stable

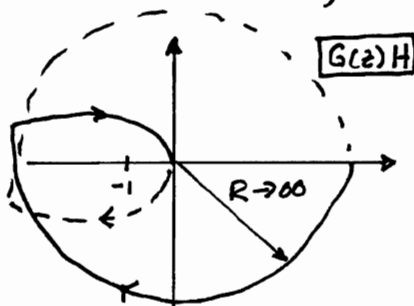
(c) Gain margin  $\approx \frac{1}{0.04} = \underline{25}$  (31.58 from Problem 7-11)

$\omega_w \approx 1.5$ ,  $\text{dB} \approx 8$ ,  $\therefore \phi_m \approx 180^\circ - 105^\circ = \underline{75^\circ}$

(d) From (c),  $K \approx \underline{25}$  (31.58)

(e) From Table P7-23,  $\omega \approx 15 \text{ rad/s}$

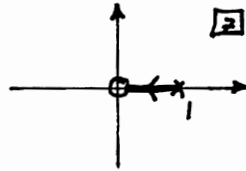
7-25.(a)



$\therefore$  always unstable.

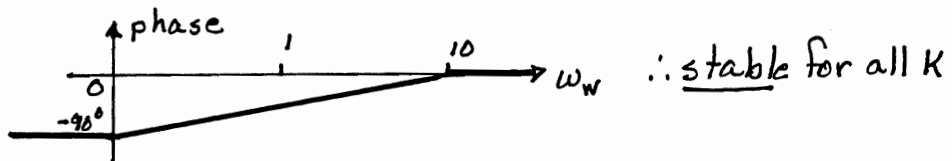
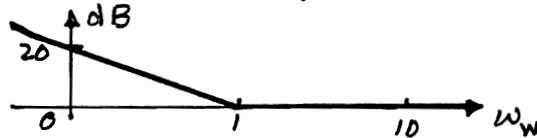
(b) Since  $K$  cannot affect the number of encirclements, the system is unstable for all  $K$ .

7-26. (a)  $D(z)G(z) = \frac{z-0.9}{z-1} \times \frac{Kz}{z-0.9} = \frac{Kz}{z-1}$



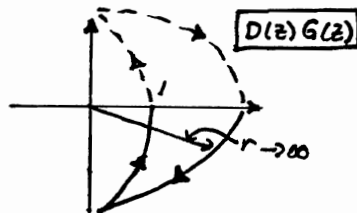
(b)  $z = \frac{1+W}{1-W}$ ,  $\therefore D(W)G(W) = \frac{K(1+W)}{1+W-1+W} = \frac{K}{2} \left( \frac{W+1}{W} \right)$

$K=2$ ,  $D(W)G(W) = \frac{1+W}{W}$



$\therefore$  stable for all  $K$

(c)



gain margin =  $\infty$   
(stable for all  $K > 0$ )  
 $\phi_m = 180^\circ$

7-27. (a) (i)  $G^*(j\omega)$ ;

$0 \leq \omega \leq \omega_s/2$

(ii)  $G(z)$ ;

$z = e^{j\omega T}$ ,  $0 \leq \omega \leq \omega_s/2$

(iii)  $G(j\omega_w)$ ;

$0 \leq \omega_w < \infty$

(iv)  $G^*(j\omega) = \frac{1}{T} \sum_{n=-\infty}^{\infty} G(j\omega + jn\omega_s)$ ;  $0 \leq \omega \leq \omega_s/2$

(b)  $G^*(j\omega) = \frac{1}{T} \sum_{n=-\infty}^{\infty} G(j\omega + jn\omega_s)$ ;  $0 \leq \omega \leq \omega_s/2$

(c) Evaluate  $G(z)$  for  $z = e^{j\omega T}$ . Also evaluate  $\omega_w = \frac{\omega}{T} \tan(\frac{\omega T}{2})$ . Then plot  $G(z)$  versus  $\omega_w$ .

(d) With sufficient numerical accuracy, the two diagrams will be the same.